E-COMMERCE ANALYTICS PROJECT

Portfolio Project Class

ABSTRACT

Comprehensive Business Intelligence & Data Analysis Capstone Project

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DEPI – Data Analysis Track

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E-COMMERCE ANALYTICS PROJECT

Comprehensive Business Intelligence & Data Analysis Capstone Project

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PROJECT OVERVIEW

Project Title: E-Commerce Business Performance Analytics & Strategic Intelligence

Duration: Six Weeks

Team Size: Already speicified

Tools Required:

- Power BI Desktop (January 2026 Version)
- Power Query for data transformation
- Power Pivot for data modeling
- DAX for calculations and measures

Dataset: FactOnlineSales and related dimension tables (DimCustomer, DimProduct, DimStore, DimDate, DimPromotion, DimCurrency)

PROJECT OBJECTIVES

This capstone project is designed to simulate a real-world business intelligence scenario where you will act as the data analytics team for an e-commerce organization. You will be responsible for:

1. Understanding the e-commerce industry landscape and competitive dynamics
2. Analyzing business performance across multiple dimensions
3. Creating actionable insights for C-level executives and managers
4. Building professional, role-specific dashboards
5. Documenting your work according to industry standards
6. Presenting findings and strategic recommendations

PART 1: INDUSTRY RESEARCH & BUSINESS CONTEXT

1.1 E-Commerce Industry Analysis

Conduct comprehensive research on the e-commerce industry covering the following areas:

1.1.1 Global Market Overview

- Global e-commerce market size and growth trends (2020-2025)
- Market projections for 2026-2030
- Regional distribution of e-commerce revenue (North America, Europe, Asia-Pacific, Middle East, Africa)
- Key drivers of e-commerce growth globally
- Impact of digital transformation and mobile commerce
- Emerging trends (social commerce, live shopping, AI personalization, etc.)

1.1.2 Egyptian E-Commerce Market

- Size and growth rate of Egyptian e-commerce market
- Market penetration and adoption rates
- Consumer behavior and preferences in Egypt
- Payment methods and digital payment adoption
- Logistics and delivery infrastructure
- Regulatory environment and e-commerce laws
- Challenges specific to the Egyptian market (internet penetration, trust, payment infrastructure)

1.1.3 Competitive Landscape

- Identify top 10 global e-commerce platforms (Amazon, Alibaba, etc.)
- Identify top 5-7 e-commerce players in Egypt and MENA region
- Market share analysis of major competitors
- Competitive positioning matrix
- Unique value propositions of market leaders
- Comparison of business models (marketplace vs. direct selling)

1.1.4 Industry Dynamics

- Porter's Five Forces analysis for e-commerce industry
- Value chain analysis
- Customer acquisition cost trends
- Customer lifetime value benchmarks
- Technology stack commonly used in e-commerce
- Omnichannel strategies and integration

1.1.5 Risks and Challenges

- Cybersecurity and data privacy concerns
- Supply chain disruptions
- Customer retention challenges
- High return rates and reverse logistics
- Price competition and margin pressure
- Regulatory compliance (GDPR, local regulations)
- Fraud and payment security
- Last-mile delivery challenges

1.1.6 Key Performance Indicators (KPIs)

Identify and define standard industry KPIs in the following categories:

Financial KPIs:

- Gross Merchandise Value (GMV)
- Revenue and Revenue Growth Rate
- Average Order Value (AOV)
- Gross Margin and Net Margin
- Customer Acquisition Cost (CAC)
- Customer Lifetime Value (CLV)
- CAC to CLV Ratio
- Return on Ad Spend (ROAS)

Operational KPIs:

- Conversion Rate
- Cart Abandonment Rate
- Order Fulfillment Rate
- Average Delivery Time
- Return Rate
- Inventory Turnover
- Stock-out Rate
- Website Traffic and Unique Visitors

Customer-Centric KPIs:

- Customer Retention Rate
- Customer Churn Rate
- Net Promoter Score (NPS)
- Repeat Purchase Rate
- Customer Satisfaction Score (CSAT)
- Active Customer Base

Marketing KPIs:

- Traffic Sources and Channel Performance
- Email Open and Click-through Rates
- Social Media Engagement Rate
- Cost Per Click (CPC) and Cost Per Acquisition (CPA)
- Promotion Effectiveness

PART 2: DATA PREPARATION & MODELING

2.1 Data Connection with Power Query

Requirements:

1. **Connect to Data Sources**
    - Import all provided tables into Power BI using appropriate connectors
    - Document the source location and connection method
2. **Data Quality Assessment**
    - Identify and document data quality issues (missing values, duplicates, inconsistencies)
    - Check for referential integrity between fact and dimension tables
    - Validate date ranges and ensure no future dates exist where inappropriate
    - Verify numeric columns for outliers or unrealistic values
3. **Data Transformation Tasks**
    - Remove duplicate records if any
    - Handle missing values with appropriate strategies (document your approach)
    - Standardize data formats (dates, currencies, text casing)
    - Create calculated columns in Power Query where appropriate:
        - Full Name in DimCustomer (FirstName + MiddleName + LastName)
        - Age calculation from BirthDate in DimCustomer
        - Age Groups (18-25, 26-35, 36-45, 46-55, 56+)
        - Customer Tenure (years since DateFirstPurchase)
        - Product Active Status based on AvailableForSaleDate and StopSaleDate
        - Promotion Active Status based on StartDate and EndDate
4. **Data Type Optimization**
    - Ensure all columns have correct data types
    - Optimize text columns (consider using whole number for IDs if appropriate)
    - Set decimal precision for currency and percentage fields
5. **Create Additional Calculated Tables**
    - Date dimension enhancements (if needed)
    - Customer segmentation table (RFM analysis preparation)
    - Product hierarchy table (Category → Subcategory → Product)

2.2 Data Modeling with Power Pivot

Requirements:

1. **Establish Relationships**
    - Create proper relationships between FactOnlineSales and all dimension tables
    - Verify relationship cardinality (1:Many, Many:1)
    - Set appropriate cross-filter directions
    - Document any bi-directional relationships and justify their use
    - Hide foreign key columns from report view
2. **Star Schema Validation**
    - Ensure the model follows star schema best practices
    - Fact table should be at the center
    - No dimension-to-dimension relationships (avoid snowflaking unless necessary)
    - Document any deviations from star schema and explain why
3. **Create Calculation Groups** (Advanced)
    - Time intelligence calculation group (YTD, MTD, QTD, Previous Year, YoY%, etc.)
    - This will demonstrate advanced DAX and modeling skills
4. **Optimize Model Performance**
    - Mark DimDate as Date Table
    - Hide unused columns
    - Consider aggregations for large fact tables (if dataset is substantial)
    - Organize fields in display folders for better user experience

2.3 DAX Measures Development

Requirements:

Create comprehensive DAX measures covering all business requirements. Organize measures into logical groups using display folders.

CRITICAL REQUIREMENT: For each major business question that requires iteration (SUMX, AVERAGEX, FILTER, etc.), you MUST:

1. Implement the measure using an iterator function
2. Implement an alternative approach using calculated columns or non-iterator functions where possible
3. Compare performance using DAX Studio Server Timings
4. Document the performance difference and CPU utilization impact
5. Provide your recommendation on which approach to use and why

Measure Categories to Create:

Sales Measures:

- Total Sales Amount
- Total Sales Quantity

- Total Return Amount
- Total Return Quantity
- Net Sales (Sales - Returns)
- Total Discount Amount
- Average Order Value
- Average Items per Order
- Sales per Customer

Profitability Measures:

- Total Cost
- Gross Profit (Sales - Cost)
- Gross Margin %
- Net Profit (after discounts and returns)
- Net Margin %
- Average Profit per Transaction

Customer Measures:

- Total Customers
- New Customers
- Returning Customers
- Customer Retention Rate
- Customer Churn Rate
- Average Customer Lifetime Value
- Active Customers (purchased in last 90 days)

Product Measures:

- Total Products Sold
- Top Selling Products
- Underperforming Products
- Product Return Rate by Product
- Average Product Price
- Price Variance %

Time Intelligence Measures:

- Sales YTD
- Sales MTD
- Sales QTD
- Previous Year Sales
- Year-over-Year Growth %
- Month-over-Month Growth %
- Same Period Last Year
- Moving Annual Total (MAT)

Promotion Effectiveness Measures:

- Sales with Promotion
- Sales without Promotion
- Promotion Uplift %
- Average Discount per Transaction
- Promotion ROI

Store Performance Measures:

- Sales by Store
- Store Productivity (Sales per SellingAreaSize)
- Top Performing Stores
- Store Contribution %

PART 3: ROLE-SPECIFIC BUSINESS QUESTIONS

For each executive role below, you must analyze their priorities and create appropriate measures, visuals, and dashboard pages to answer their specific questions.

3.1 CHIEF EXECUTIVE OFFICER (CEO) QUESTIONS

Strategic Focus: Overall business performance, growth trajectory, market position, and long-term sustainability

Business Questions:

1. **Overall Performance:**
    - What is our total Gross Merchandise Value (GMV) and how has it evolved over time?
    - What is our year-over-year revenue growth rate?
    - Are we meeting our quarterly and annual revenue targets?
    - What is our current profitability and profit margin trend?
2. **Growth Analysis:**
    - Which geographic regions are showing the strongest growth?
    - What is our customer base growth rate?
    - Which product categories are driving our growth?
    - What is our market penetration rate in different regions?
3. **Customer Intelligence:**
    - What is our customer acquisition trend?
    - What is our customer retention rate and how does it compare to industry benchmarks?

    - What is the average customer lifetime value?
    - What percentage of revenue comes from repeat customers?
4. **Strategic Positioning:**
    - What are our top 10 products by revenue and profitability?
    - Which customer segments are most valuable to our business?
    - What is our average order value trend?
    - How effective are our promotional campaigns in driving incremental revenue?
5. **Risk and Opportunities:**
    - What is our product return rate and is it increasing or decreasing?
    - Are there any concerning trends in customer churn?
    - Which stores or regions are underperforming?
    - What seasonal patterns should we prepare for?
6. **Operational Excellence:**
    - What is our sales per employee (if employee data is available via stores)?
    - How efficiently are we converting traffic to sales?
    - What is our inventory turnover rate by category?

3.2 CHIEF FINANCIAL OFFICER (CFO) QUESTIONS

Strategic Focus: Financial health, profitability, cost management, revenue optimization, and financial forecasting

Business Questions:

1. **Revenue Analysis:**
    - What is the monthly, quarterly, and annual revenue trend?
    - What is the revenue breakdown by product category, region, and customer segment?
    - What is our revenue concentration (what % comes from top 20% of customers)?
    - How predictable is our revenue (coefficient of variation)?
2. **Profitability Analysis:**
    - What is our gross margin by product category and individual product?
    - What is our net margin after accounting for discounts and returns?
    - Which products or categories are most and least profitable?
    - What is the contribution margin by customer segment?
3. **Cost Management:**
    - What is our total cost of goods sold (COGS) and its trend?
    - What is the cost as a percentage of revenue?
    - How do unit costs vary across products and suppliers?
    - What is the impact of returns on our bottom line?
4. **Pricing and Discounting:**
    - What is our average selling price vs. unit cost by category?
    - What is our price realization (actual selling price vs. listed price)?
    - How much revenue are we losing to discounts?

    - What is the discount % trend and its impact on margins?
    - Are our promotional discounts driving profitable volume?
5. **Customer Economics:**
    - What is our customer acquisition cost by channel?
    - What is the average customer lifetime value?
    - What is the CAC to CLV ratio?
    - What is the payback period for customer acquisition?
6. **Cash Flow Indicators:**
    - What is our average days sales outstanding (based on sales patterns)?
    - What is our inventory value and turnover rate?
    - What are the working capital implications of returns?
7. **Currency Impact:**
    - How are different currency transactions performing?
    - What is our exposure to currency fluctuations?
    - Which currencies generate the most revenue and profit?
8. **Financial Forecasting:**
    - Based on trends, what is the projected revenue for next quarter?
    - What is the projected annual revenue at current growth rate?
    - What scenarios could impact our financial targets?

3.3 CHIEF OPERATING OFFICER (COO) QUESTIONS

Strategic Focus: Operational efficiency, process optimization, fulfillment excellence, and resource utilization

Business Questions:

1. **Order Fulfillment:**
    - What is our average order processing time by region and store?
    - What is our order fulfillment rate (orders shipped vs. orders received)?
    - How many orders are being processed daily, weekly, monthly?
    - What is our peak order volume period and how well do we handle it?
2. **Inventory Management:**
    - What is our inventory turnover ratio by product category?
    - Which products are fast-moving vs. slow-moving?
    - What is our stock-out frequency?
    - What is the optimal inventory level by product?
3. **Returns Management:**
    - What is our overall return rate?
    - Which products have the highest return rates?
    - What is the return rate by customer segment?
    - What are the patterns in return reasons (if data available)?
    - What is the cost impact of returns on operations?
4. **Store Performance:**

    - Which stores are most productive (sales per square foot)?
    - What is the sales distribution across stores?
    - How do store types perform relative to each other?
    - Are there stores consistently underperforming?
5. **Operational Efficiency:**
    - What is our average items per order?
    - What is the order value distribution (% of small vs. large orders)?
    - How efficiently are we utilizing our store selling area?
    - What is the relationship between store size and performance?
6. **Process Optimization:**
    - What are our peak vs. off-peak sales periods?
    - How should we allocate resources based on demand patterns?
    - Which operational processes are bottlenecks?
    - What is the day-of-week and time-of-day sales pattern?
7. **Quality Metrics:**
    - What is the defect rate (returns as a proxy)?
    - Are there quality issues with specific suppliers or manufacturers?
    - What is the customer satisfaction trend?
8. **Capacity Planning:**
    - What is our current capacity utilization?
    - Do we have sufficient capacity for projected growth?
    - Which regions require capacity expansion?

3.4 CHIEF SALES OFFICER QUESTIONS

Strategic Focus: Sales performance, sales team productivity, channel effectiveness, and revenue growth strategies

Business Questions:

1. **Sales Performance:**
    - What is our overall sales trend (daily, weekly, monthly, quarterly)?
    - What is our sales growth rate by period?
    - Are we meeting our sales targets and quotas?
    - What is the sales performance by region and store?
2. **Sales Channel Analysis:**
    - How are different stores performing in terms of sales volume?
    - Which stores contribute most to total revenue?
    - What is the sales per store and sales per square foot?
    - Should we expand, maintain, or close any stores?
3. **Product Sales Mix:**
    - What are our top-selling products by volume and revenue?
    - What is the sales distribution across product categories?
    - Which products have the highest and lowest sales velocity?

    - What is the product penetration rate across customer base?
4. **Customer Acquisition and Conversion:**
    - How many new customers are we acquiring monthly?
    - What is the conversion rate from first-time to repeat buyer?
    - What is the average time to second purchase?
    - Which customer segments have the highest conversion rates?
5. **Sales Productivity:**
    - What is the average transaction value?
    - What is the average items per transaction?
    - How can we increase basket size?
    - What is the cross-sell and up-sell effectiveness?
6. **Territory and Geographic Performance:**
    - Which geographic areas are high-performing vs. underperforming?
    - Where should we focus our sales efforts?
    - Are there untapped markets or territories?
    - What is the customer density by geography?
7. **Seasonal and Trend Analysis:**
    - What are our seasonal sales patterns?
    - Which products are seasonal and which are year-round?
    - How should we plan inventory for seasonal demand?
    - What is the impact of holidays on sales?
8. **Sales Forecasting:**
    - What is the projected sales for next month, quarter, and year?
    - What is the sales pipeline and expected conversion?
    - What factors are most predictive of sales performance?
9. **Customer Behavior:**
    - What is the repeat purchase rate?
    - What is the average time between purchases?
    - Which customers are at risk of churning?
    - What is the customer purchase frequency distribution?

3.5 CHIEF MARKETING OFFICER (CMO) QUESTIONS

Strategic Focus: Marketing ROI, campaign effectiveness, customer segmentation, brand performance, and customer engagement

Business Questions:

1. **Campaign Performance:**
    - What is the overall effectiveness of our promotional campaigns?
    - Which promotions generated the highest revenue?
    - What is the ROI of each promotional campaign?
    - What is the uplift in sales during promotion periods vs. non-promotion periods?
    - What is the average discount percentage by promotion type?

2. **Customer Segmentation:**
    - What are our key customer segments (by demographics, behavior, value)?
    - Which customer segments are most profitable?
    - What is the size and growth rate of each segment?
    - How do different segments respond to marketing initiatives?
    - What is the lifetime value by customer segment?
3. **Customer Acquisition:**
    - What is our customer acquisition cost by source/channel?
    - Which marketing channels are most effective for acquisition?
    - What is the quality of customers acquired from different channels?
    - What is the new customer growth trend?
4. **Customer Retention and Loyalty:**
    - What is our customer retention rate by cohort?
    - What is the repeat purchase rate?
    - How many customers make 1, 2, 3, 4+ purchases?
    - What drives customer loyalty in our business?
    - What is the customer churn rate and what are the early warning signs?
5. **Product Marketing:**
    - Which products have the highest brand awareness (sales volume as proxy)?
    - What is the performance of new product launches?
    - Which product categories need more marketing support?
    - What is the product affinity (which products are bought together)?
6. **Brand Performance:**
    - How are different brands (BrandName) performing in terms of sales and margin?
    - What is the brand market share within our product portfolio?
    - Which brands command premium pricing?
    - What is the brand loyalty (repeat purchases of same brand)?
7. **Pricing Strategy:**
    - What is the price elasticity of demand for our products?
    - How do customers respond to discounts?
    - What is the optimal discount level to drive volume without eroding margin?
    - Are we pricing competitively?
8. **Customer Demographics:**
    - What is our customer profile (age, gender, income, education)?
    - How do different demographic groups perform?
    - Which demographics have the highest lifetime value?
    - Are we reaching our target demographic?
9. **Customer Journey:**
    - What is the average customer journey from first to repeat purchase?
    - How many touchpoints before purchase?
    - What is the drop-off rate at each stage?
10. **Marketing Budget Allocation:**
    - How should we allocate marketing budget across channels?
    - What is the marginal return on marketing spend?
    - Which initiatives deserve more investment?

3.6 PRODUCT MANAGER QUESTIONS

Strategic Focus: Product performance, product lifecycle management, portfolio optimization, and product development insights

Business Questions:

1. **Product Performance:**
    - What is the sales performance of each product?
    - Which products are top performers and which are underperforming?
    - What is the revenue contribution by product?
    - What is the profitability by product (margin analysis)?
2. **Product Portfolio Analysis:**
    - How many active products do we have?
    - What is the sales concentration (how many products drive 80% of revenue)?
    - Which products should we discontinue?
    - Where are the gaps in our product portfolio?
3. **Product Category Insights:**
    - How are different product categories and subcategories performing?
    - What is the category mix trend?
    - Which categories are growing and which are declining?
    - What is the profitability by category?
4. **Product Attributes Analysis:**
    - How do different product attributes affect sales (size, color, style, class)?
    - Which colors/sizes/styles are most popular?
    - Are there underserved attribute combinations?
    - What is the sales distribution by product attributes?
5. **Product Lifecycle:**
    - What is the sales trend by product from launch to current?
    - Which products are in growth, maturity, or decline stage?
    - What is the average product lifecycle in our portfolio?
    - When should we phase out products?
6. **Product Returns and Quality:**
    - Which products have the highest return rates?
    - Is there a quality issue with specific products?
    - What is the correlation between product attributes and returns?
    - Which manufacturers/brands have the highest quality (lowest returns)?
7. **Product Pricing:**
    - What is the price positioning of our products?
    - Are we pricing optimally for each product?
    - What is the price-volume relationship?
    - Which products can support price increases?
8. **Inventory Planning:**
    - Which products have the fastest turnover?

    - Which products are slow-moving and tying up capital?
    - What is the optimal stock level by product?
    - Which products frequently stock out?
9. **Product Bundling and Cross-Sell:**
    - Which products are frequently bought together?
    - What bundling opportunities exist?
    - What is the cross-sell success rate?
10. **Competitive Product Analysis:**
    - How are our products positioned vs. competitors?
    - What is our product mix compared to industry standards?
    - Where should we invest in product development?
11. **Supplier and Manufacturer Performance:**
    - Which manufacturers deliver the best-selling products?
    - What is the profitability by manufacturer?
    - Are there supplier concentration risks?

PART 4: DASHBOARD DEVELOPMENT

4.1 Dashboard Requirements

Develop professional, interactive dashboards that address the business questions outlined in Part 3. You must create:

4.1.1 Executive Dashboard (Strategic Level)

Primary Audience: CEO, Board of Directors

Requirements:

- High-level KPIs and metrics
- Year-over-year performance comparison
- Key trends and insights
- Strategic alerts and warnings
- Clean, minimal design with focus on insights
- Drill-through capabilities to operational dashboards

Must Include:

- Total revenue, profit, and margin
- Customer acquisition and retention metrics
- Growth trends
- Top/bottom performers
- Geographic performance map
- Quarterly and annual performance vs. targets

4.1.2 Financial Dashboard

Primary Audience: CFO, Finance Team

Requirements:

- Detailed financial metrics
- Profitability analysis
- Cost analysis
- Revenue breakdown
- Margin analysis by multiple dimensions
- Financial forecasting visuals

Must Include:

- P&L summary view
- Revenue waterfall
- Margin analysis by product/category/region
- Cost breakdown
- Discount impact analysis
- Currency performance
- Month-over-month and YoY comparisons

4.1.3 Operations Dashboard

Primary Audience: COO, Operations Managers

Requirements:

- Operational efficiency metrics
- Fulfillment performance
- Inventory metrics
- Store performance
- Returns analysis
- Process efficiency indicators

Must Include:

- Order volume trends
- Fulfillment metrics
- Return rate analysis
- Store productivity matrix
- Inventory turnover by category
- Peak period analysis

4.1.4 Sales Dashboard

Primary Audience: Chief Sales Officer, Sales Managers

Requirements:

- Sales performance by multiple dimensions
- Sales trends and forecasts
- Territory/store performance
- Product sales mix
- Sales productivity metrics

Must Include:

- Sales trend with forecast
- Sales by store/region
- Top products
- Sales funnel/conversion
- Customer acquisition trend
- Sales target vs. actual

4.1.5 Marketing Dashboard

Primary Audience: CMO, Marketing Team

Requirements:

- Campaign performance
- Customer segmentation analysis
- Promotion effectiveness
- Customer behavior insights
- Brand performance

Must Include:

- Promotion ROI analysis
- Customer segmentation matrix
- Customer lifetime value analysis
- Repeat purchase behavior
- Brand performance comparison
- Customer demographic analysis

4.1.6 Product Performance Dashboard

Primary Audience: Product Managers

Requirements:

- Product-level performance metrics
- Portfolio analysis
- Product lifecycle view
- Attribute analysis
- Return analysis by product

Must Include:

- Product performance matrix
- Category/subcategory analysis
- Product attributes performance
- Product profitability analysis
- Product return rate analysis
- Inventory turnover by product

4.2 Dashboard Design Principles

Your dashboards must follow these design principles:

1. **User-Centric Design:**
    - Design for the specific role and their decision-making needs
    - Prioritize most important metrics at the top
    - Use appropriate visual hierarchy
2. **Visual Best Practices:**
    - Use appropriate chart types for data (bar, line, scatter, map, etc.)
    - Avoid chart junk and unnecessary decoration
    - Use consistent color scheme aligned with brand
    - Ensure charts are readable and not cluttered
    - Proper axis labels, titles, and legends
3. **Interactivity:**
    - Implement slicers for key dimensions (date, product, region, store)
    - Cross-filtering between visuals
    - Drill-through to detailed pages
    - Tooltips with additional context
    - Bookmarks for different views/scenarios
4. **Performance:**
    - Optimize DAX measures for performance
    - Use appropriate visual types (avoid custom visuals unless necessary)
    - Implement incremental refresh if dataset is large
    - Test dashboard performance and loading times
5. **Accessibility:**
    - Use color-blind friendly palettes
    - Ensure sufficient contrast
    - Provide alt text for visuals
    - Tab order for keyboard navigation

4.3 Dashboard Navigation

Create a navigation structure:

- Home/Landing page with navigation to all dashboards
- Consistent navigation bar/menu across all pages
- Breadcrumb navigation
- Back buttons
- Drill-through return navigation

PART 5: PERFORMANCE OPTIMIZATION & ITERATOR ANALYSIS

5.1 DAX Performance Analysis - MANDATORY REQUIREMENT

For the following scenarios, you MUST implement both iterator and non-iterator approaches and analyze performance:

Scenario 1: Total Sales Calculation

Requirement: Calculate total sales where Sales Amount = Quantity × Unit Price

- **Approach A:** Use SUMX to iterate through rows
- **Approach B:** Create calculated column in Power Query, then use SUM
- **Analysis Required:**
  - Compare execution times using DAX Studio
  - Analyze Storage Engine vs Formula Engine query time
  - Document CPU utilization difference
  - Make recommendation with justification

Scenario 2: Customer Lifetime Value

Requirement: Calculate total sales per customer

- **Approach A:** Use SUMX with FILTER
- **Approach B:** Use CALCULATE with ALL/REMOVEFILTERS
- **Analysis Required:**
  - Performance comparison
  - Memory consumption analysis
  - Recommendation for production use

Scenario 3: Average Order Value with Conditions

Requirement: Calculate average order value excluding returns

- **Approach A:** Use AVERAGEX with FILTER
- **Approach B:** Use DIVIDE with SUM measures
- **Analysis Required:**
  - Performance metrics
  - Accuracy validation
  - Best practice recommendation

Scenario 4: Product Performance Ranking

Requirement: Rank products by sales

- **Approach A:** Use RANKX
- **Approach B:** Use calculated column approach
- **Analysis Required:**
  - Performance at different data volumes
  - Real-time vs. cached performance
  - Use case recommendations

5.2 Performance Documentation Requirements

For each scenario above, document:

1. **DAX Code:** Provide both versions of the measure
2. **DAX Studio Results:** Screenshot of Server Timings pane showing:
   - Total execution time
   - Storage Engine time
   - Formula Engine time
   - Number of Storage Engine queries
3. **Performance Metrics Table:**
   - Execution time (ms)
   - SE vs FE percentage
   - CPU utilization pattern
4. **Analysis:**
   - Why the performance difference exists
   - When to use each approach
   - Scalability considerations
5. **Recommendation:**
   - Which approach to use in production
   - Justification based on data

5.3 General Performance Optimization

Document your optimization strategies:

- Use of variables in DAX
- Avoiding circular references
- Reducing cardinality where possible
- Proper use of CALCULATE vs. FILTER
- Minimizing iterator use when unnecessary
- Column vs. measure decision framework

PART 6: PROJECT DOCUMENTATION

6.1 Documentation Requirements

Create comprehensive project documentation following professional standards:

6.1.1 Executive Summary

- Project overview and objectives
- Scope and deliverables
- Key findings summary
- High-level recommendations

6.1.2 Industry Research Report

- Consolidate all findings from Part 1
- Include citations and sources
- Visual representations (charts, graphs)
- SWOT analysis for the e-commerce industry
- Competitive landscape summary

6.1.3 Data Dictionary

Create a complete data dictionary including:

For Each Table:

- Table name and description
- Table type (Fact/Dimension)
- Number of rows
- Grain/level of detail

For Each Column:

- Column name
- Data type
- Description/business meaning

- Sample values
- Null allowed? (Yes/No)
- Key type (Primary, Foreign, None)

Example Format:

Table	Column	Data Type	Description	Sample Values	Nullable	Key Type
FactOnlineSales	OnlineSalesKey	Int	Unique identifier for each sale transaction	1, 2, 3...	No	Primary

6.1.4 Data Model Documentation

Document your data model:

- Entity Relationship Diagram (ERD)
- Relationship descriptions (which tables are related and why)
- Cardinality explanation
- Filter direction rationale
- Any calculated tables and their purpose

6.1.5 Business Glossary

Define all business terms, KPIs, and metrics used in the project:

Format:

- **Term:** Customer Lifetime Value (CLV)
- **Definition:** The total revenue expected from a customer over their entire relationship with the business
- **Formula:** Sum of all purchases by customer
- **Business Use:** Helps identify high-value customers for retention efforts
- **Related Metrics:** Customer Acquisition Cost, Retention Rate

Create definitions for ALL metrics used in your dashboards.

6.1.6 DAX Measures Documentation

Document all DAX measures:

Format for each measure:

- **Measure Name:** Total Sales
- **Display Folder:** Sales Metrics
- **Description:** Sum of all sales amounts excluding returns
- **Formula:** [provide DAX code]

- **Dependencies:** [list any measures this depends on]
- **Used In:** [which dashboards/visuals use this]
- **Performance Notes:** [any relevant performance information]

6.1.7 Dashboard User Guide

For each dashboard, document:

- Purpose and primary audience
- How to use interactivity features
- Explanation of each visual
- How to interpret the metrics
- Common questions answered by this dashboard

6.1.8 Technical Documentation

Include:

- Power BI Desktop version used
- Data refresh schedule recommendations
- Performance optimization implemented
- Known limitations
- Future enhancement recommendations
- Deployment instructions

6.1.9 Assumptions and Limitations

Document:

- Any assumptions made during analysis
- Data quality issues encountered
- Limitations of the analysis
- Missing data or information gaps
- Constraints that affected the project

PART 7: INSIGHTS & RECOMMENDATIONS

7.1 Analysis and Findings

Based on your analysis, document:

7.1.1 Key Findings

- Top 10 most important insights discovered

- Trends identified
- Anomalies or unexpected patterns
- Performance gaps
- Opportunities identified

7.1.2 Segment Analysis

- Customer segment insights
- Product segment insights
- Geographic insights
- Temporal patterns

7.1.3 Performance Assessment

- What's working well
- What's underperforming
- Comparative benchmarks
- Gap analysis

7.2 Strategic Recommendations

Provide actionable recommendations for each stakeholder:

7.2.1 For CEO

- Strategic direction recommendations
- Growth opportunities
- Risk mitigation strategies
- Investment priorities

7.2.2 For CFO

- Profitability improvement opportunities
- Cost optimization recommendations
- Pricing strategy suggestions
- Financial risk management

7.2.3 For COO

- Operational efficiency improvements
- Process optimization recommendations
- Resource allocation suggestions
- Supply chain enhancements

7.2.4 For Chief Sales Officer

- Sales strategy recommendations
- Territory/channel optimization
- Sales productivity improvements
- Customer conversion strategies

7.2.5 For CMO

- Marketing strategy recommendations
- Campaign optimization
- Customer retention initiatives
- Segmentation and targeting refinements

7.2.6 For Product Manager

- Product portfolio optimization
- New product opportunities
- Product discontinuation recommendations
- Pricing adjustments

7.3 Implementation Roadmap

Provide a prioritized roadmap:

- Quick wins (immediate actions with high impact)
- Short-term initiatives (1-3 months)
- Medium-term initiatives (3-6 months)
- Long-term strategic initiatives (6-12 months)

PART 8: PRESENTATION

8.1 Final Presentation Requirements

Prepare a professional presentation (PowerPoint/Google Slides) covering:

Presentation Structure:

1. **Title Slide**
    - Project title
    - Team members
    - Date
    - Course information
2. **Executive Summary** (2-3 slides)
    - Project objectives
    - Scope

    - Key deliverables
    - High-level findings
3. **Industry Overview** (3-4 slides)
    - E-commerce market landscape
    - Competitive environment
    - Key challenges and opportunities
    - Industry KPIs
4. **Data & Methodology** (2-3 slides)
    - Data sources and scope
    - Data model overview
    - Analytical approach
    - Tools and techniques used
5. **Key Insights** (5-7 slides)
    - Most important findings
    - Supporting data visualizations
    - Trends and patterns
    - Performance highlights and concerns
6. **Dashboard Demo** (3-5 slides)
    - Screenshots of key dashboards
    - Highlight interactivity features
    - Show how it answers business questions
    - User experience highlights
7. **Performance Analysis** (2-3 slides)
    - DAX performance comparison results
    - Iterator vs non-iterator analysis
    - Optimization strategies implemented
    - Performance recommendations
8. **Recommendations** (4-5 slides)
    - Strategic recommendations by stakeholder
    - Prioritized action items
    - Implementation roadmap
    - Expected impact
9. **Challenges & Learnings** (1-2 slides)
    - Challenges faced during the project
    - How they were overcome
    - Key learnings
    - What would you do differently
10. **Q&A** (1 slide)
    - Questions and discussion

8.2 Presentation Delivery

- **Duration:** 20-30 minutes + 10 minutes Q&A
- **Format:** Live demo of Power BI dashboards + slide presentation
- **Preparation:** Practice as a team, ensure smooth transitions
- **Technical:** Test all technology before presentation, have backup plan

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PART 9: EVALUATION CRITERIA

Your project will be evaluated on the following criteria:

9.1 Industry Research (15 points)

- Depth and breadth of research
- Quality of sources
- Relevance to project
- Critical analysis
- Professional presentation of findings

9.2 Data Preparation & Modeling (20 points)

- Data quality assessment and cleaning
- Power Query transformations
- Data model design (star schema, relationships)
- Optimization and best practices
- Documentation quality

9.3 DAX Development (20 points)

- Correctness of measures
- Complexity and sophistication
- Performance optimization
- Iterator vs non-iterator analysis
- Code organization and documentation
- Use of advanced DAX concepts

9.4 Dashboard Development (20 points)

- Addresses all business questions
- Visual design and aesthetics
- Interactivity and user experience
- Appropriate chart types
- Performance
- Navigation and usability

9.5 Documentation (15 points)

- Completeness
- Clarity and professionalism
- Data dictionary quality

- Business glossary
- Technical documentation
- Adherence to standards

9.6 Insights & Recommendations (10 points)

- Quality of insights
- Actionability of recommendations
- Business acumen
- Strategic thinking
- Justification and supporting evidence

9.7 Presentation (10 points)

- Clarity and structure
- Professionalism
- Demonstration quality
- Team coordination
- Time management
- Q&A handling

9.8 Bonus Points (up to 10 points)

- Innovation and creativity
- Going beyond requirements
- Advanced techniques
- Exceptional quality
- Business impact potential

Total: 110 points possible

PART 10: SUBMISSION REQUIREMENTS

10.1 Deliverables Checklist

Submit the following:

10.1.1 Power BI File

- [ ] .pbix file with all dashboards
- [ ] File size optimized
- [ ] All relationships configured
- [ ] All measures organized in display folders

- [ ] Navigation fully functional
- [ ] File name: TeamName_ECommerce_Analytics.pbix

10.1.2 Documentation Package

- [ ] Executive Summary (PDF)
- [ ] Industry Research Report (PDF)
- [ ] Data Dictionary (Excel/PDF)
- [ ] Business Glossary (Excel/PDF)
- [ ] Data Model Documentation (PDF with ERD)
- [ ] DAX Measures Documentation (Excel/PDF)
- [ ] Performance Analysis Report (PDF with DAX Studio screenshots)
- [ ] Dashboard User Guide (PDF)
- [ ] Technical Documentation (PDF)
- [ ] Insights & Recommendations Report (PDF)

10.1.3 Presentation

- [ ] PowerPoint/Google Slides presentation
- [ ] Presentation notes/script
- [ ] Any supporting materials

10.1.4 Supporting Files

- [ ] Power Query M code documentation (if extensive transformations)
- [ ] DAX Studio performance test results (Excel)
- [ ] Any external data sources used (with documentation)

10.2 Submission Format

- Create a compressed folder (.zip) named: TeamName_ECommerce_Project.zip
- Include all deliverables organized in subfolders:
    - /PowerBI (contains .pbix file)
    - /Documentation (all PDF documentation)
    - /Presentation (PowerPoint and supporting materials)
    - /Supporting (any additional files)

10.3 Submission Deadline

Due Date: Will be scheduled on phases – announced separately

Submission Method: Email me at waleed.mouhammed@icloud.com

Late Submission Policy: 2% of the whole project grade will be deducted for each day after the intended sumbission date

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PART 11: RESOURCES AND SUPPORT

11.1 Recommended Resources

Power BI Learning:

- Microsoft Power BI Documentation: https://docs.microsoft.com/power-bi/
- DAX Guide: https://dax.guide/
- SQLBI (Marco Russo & Alberto Ferrari): https://www.sqlbi.com/
- Guy in a Cube (YouTube): Power BI tutorials

E-Commerce Industry:

- Statista: E-commerce statistics
- eMarketer: Digital commerce insights
- World Bank: Global e-commerce data
- Egypt Ministry of Communications and Information Technology reports

DAX Performance:

- DAX Studio: https://daxstudio.org/
- SQLBI Performance Optimization articles
- Power BI Performance Best Practices (Microsoft Docs)

Design Inspiration:

- Power BI Community Gallery
- Tableau Public
- Data visualization best practices (Edward Tufte, Stephen Few)

11.2 Getting Help

- Office hours: I will respond to your project related questions within 24 H
- Discussion forum: WhatsApp
- Team collaboration expectations: This is a team project, any team will not work together will be excluded from the competition

11.3 Academic Integrity

- All work must be original – AI is allowed with a precent NOT greater than 25% of work done.
- Properly cite all sources
- Each team member must contribute meaningfully

- Include individual contribution statement if required

PART 12: PROJECT TIMELINE (SUGGESTED)

Week 1:

- Conduct industry research
- Familiarize with dataset
- Initial data exploration

Week 2:

- Data quality assessment
- Begin Power Query transformations
- Create data dictionary
- Finalize data model
- Establish relationships

Week 3:

- Begin DAX measure development
- Iterator performance analysis
- Start dashboard wireframes
- Begin dashboard development
- Executive dashboard

Week 4:

- Complete all dashboards
- Finalize interactivity
- Performance optimization
- Begin documentation

Week 5:

- Complete all dashboards
- Finalize interactivity
- Performance optimization
- Begin documentation

Week 6:

- Finalize presentation
- Practice presentation

- Final quality check
- Submit project

CONCLUSION

This project is designed to simulate a real-world business intelligence engagement. Approach it with professionalism, analytical rigor, and creativity. This is your opportunity to demonstrate mastery of Power BI, DAX, data modeling, business analysis, and communication skills.

Your work should be of portfolio quality—something you would be proud to show to a potential employer. Take pride in your analysis, be thorough in your documentation, and deliver insights that could genuinely help an e-commerce business succeed.

Good luck, and may your data tell a compelling story!

Project Author: Waleed Mouhammed

Course: Data Analytics with Power BI

Institution: DEPI

Date: February 2026